

THANH TRINH-XUAN

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EDUCATION

Hanoi University of Science and Technology (HUST) <i>Bachelor of Engineering in Electrical Engineering</i> • GPA: 3.2/4.0 • Research interests: mathematical programming and distributed algorithms in power systems, peer-to-peer energy markets in multi-microgrids, optimization and operation control in resilience-oriented interconnected smart grids • Affiliation: Member of 100 Renewable Energy Laboratory. • Bachelor's Thesis: Resilience-Oriented Peer-to-Peer Energy Trading in Multi-Microgrids via Network-Constrained Distributed Predictive Control • Relevant Coursework: Power System Analysis, Power System Automation, Power System Protection and Control, Digital System Design, Programming Techniques.	Hanoi, Vietnam <i>Aug. 2026 (Expected)</i>
Lam Son High School for the Gifted Student <i>High School Diploma, Mathematics Specialization</i>	Thanh Hoa, Vietnam <i>Jul. 2022</i>

RESEARCH EXPERIENCE

Undergraduate Researcher <i>Power Grid and Renewable Energy Laboratory, Hanoi, Vietnam</i> <i>Supervisor: Assoc. Prof. Nguyen Duc Tuyen</i>	<i>Sep. 2024 – Present</i>
Bachelor's Thesis • Modeled P-Q-V dynamics using Second-Order Cone Programming (SOCP) and integrated load shedding penalty functions to mitigate relaxation inexactness (fictitious currents), ensuring exact grid constraint enforcement. • Implemented Analytical Target Cascading (ATC) to maintain decentralized coordination when extreme faults cause network fragmentation and render centralized control unviable. • Designed Model Predictive Control (MPC) frameworks for dynamic rescheduling to mitigate immediate power imbalances during blackouts that invalidate static day-ahead dispatch.	<i>2025 – 2026</i>
Project: Distributed Energy Management Frameworks • Developed a 3-layer architecture linking MILP for discrete unit commitment, ADMM for decentralizing tie-lines without exposing private data, and Nucleolus for fair Peer-to-Peer (P2P) profit sharing. • Applied Demand Side Management (DSM) to optimized local microgrid load scheduling for mitigating peak-load stress and reducing operational costs during islanded mode. • Implemented an adaptive penalty tuning mechanism based on primal-dual residual balancing to resolve ADMM divergence caused by constrained P2P parameters, ensuring rapid convergence.	<i>2024 – 2025</i>
Lab Contribution & Mentorship • Architected custom Multi-Agent AI workflows to accelerate personal literature synthesis. • Mentored junior researchers in academic reading methodologies and algorithmic modeling, driving lab research productivity.	<i>2024 – Present</i>

ACADEMIC PROJECTS

Machine Learning for EV Load Forecasting

Mar. 2026 – April 2026

- Engineered a Proxy-Lag Cascade Ensemble architecture utilizing dual XGBoost models ($t+24h$ and $t+1$) to eliminate phase lag and accurately forecast peak EV load demand.

Industrial Power Supply & Distribution Network Design

Oct. 2025 – Dec. 2025

- Designed a high/low-voltage distribution network for a large-scale manufacturing plant, optimizing the techno-economic balance between CAPEX and OPEX without compromising system stability.
- Sized transformers, optimized cable cross sections, and performed short circuit calculations to prevent thermal overloads and ensure reliability across all operating conditions.

Electronic & Hardware Prototyping

2023 – 2024

- Designed and physically prototyped an AC current measurement circuit and a DC-DC Buck converter to ensure stable voltage regulation and high-precision monitoring for hardware testing.
- Simulated layouts using LTspice, Proteus, and MATLAB/Simulink to validate circuit robustness prior to physical fabrication.

PUBLICATIONS & PRESENTATIONS

- [1] **T. X. Thanh (1st Author)**, N. T. Anh, T. X. Hung, N. D. Tuyen, T. T. Son, G. Fujita, "A Decentralized MILP-ADMM Framework for P2P Energy Trading in Interconnected Microgrids," *10th International Conference on Green Energy and Applications (ICGEA)*, Mar. 2026. **[Oral Presentation]**
- [2] L. A. Quan, **T. X. Thanh (2nd Author)**, N. T. Anh, P. T. Vinh, T. X. Hung, N. D. Tuyen, T. T. Son, "Microgrid Optimization with MILP-based Demand Side Management," *Journal of Science and Technology (JST)*, vol. 36, Jun. 2026. **[Oral Presentation at Vietnam Student Forum]**

SKILLS

Optimization: MILP, SOCP, Convex Relaxation

Control & Algorithms: ADMM, ATC, MPC, Game Theory

Programming: Python (Pyomo), C, C++

EE Simulation: Simulink, LTspice, PVsyst, Proteus

Tools & AI: Multi-Agent AI, LaTeX, Git

Languages: English (IELTS 6.0, Target 7.0)

HONORS & AWARDS

2026 Best Presentation Award, 10th Int. Conf. on Green Energy and Applications (ICGEA)

REFERENCES

Assoc. Prof. Nguyen Duc Tuyen

Associate Professor and Head of PGRE Lab, Power Systems Department, HUST, Vietnam

Adjunct Assoc. Prof, Shibaura Institute of Technology, Japan

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